

## 22.4 Variance-stabilizing transformations

For many responses the variance is tied to the mean by the nature of the measurement: counts, proportions, amounts measured with a fixed *relative* precision. Then  $\text{Var}(Y) = \phi v(\mu)$  for a known or plausible variance function  $v$ , where  $\mu = E(Y)$  and  $\phi$  is a constant. The transformation that makes the variance constant turns out to be determined by  $v$  up to an affine change.

### 22.4.1 The idea

If  $g$  is smooth and  $Y$  is concentrated near  $\mu$ , then  $g(Y) \approx g(\mu) + g'(\mu)(Y - \mu)$ , so

$$\text{Var } g(Y) \approx g'(\mu)^2 \phi v(\mu).$$

This is constant in  $\mu$  exactly when  $g'(\mu)$  is proportional to  $v(\mu)^{-1/2}$ , that is, when

$$g(\mu) = \int^\mu \frac{dt}{\sqrt{v(t)}}. \quad (22.4.1)$$

Part (a) of the theorem below makes this precise, with  $m$  measuring the information; it is Exercise (B1), where Fisher's  $z$  arose this way. Part (b) adds uniqueness.

#### Theorem 22.4.1 — Variance-stabilizing transformations

Let  $I$  be an open interval and  $v : I \rightarrow (0, \infty)$  continuous. Suppose that for every  $\theta \in I$  the random variables  $T_m$  satisfy  $\sqrt{m}(T_m - \theta) \rightarrow N(0, v(\theta))$  in distribution as  $m \rightarrow \infty$ . Fix  $t_0 \in I$  and let  $g(t) = \int_{t_0}^t v(s)^{-1/2} ds$  for  $t \in I$ , extended in any way outside  $I$ .

- (a) For every  $\theta \in I$ ,  $\sqrt{m}\{g(T_m) - g(\theta)\} \rightarrow N(0, 1)$  in distribution.
- (b) If  $h : I \rightarrow \mathbb{R}$  is continuously differentiable and  $\sqrt{m}\{h(T_m) - h(\theta)\} \rightarrow N(0, 1)$  for every  $\theta \in I$ , then  $h = g + c$  or  $h = -g + c$  on  $I$  for a constant  $c$ .

*Proof.*

- (a) By the fundamental theorem of calculus  $g' = v^{-1/2}$  on  $I$ , so this is Exercise (B1) with  $s = v^{1/2}$ ; the values of  $g$  outside  $I$  do not matter because  $T_m \rightarrow \theta$  in probability.
- (b) By the delta method (stated in [Section 14.4](#)),  $\sqrt{m}\{h(T_m) - h(\theta)\} \rightarrow N(0, h'(\theta)^2 v(\theta))$ . Limits in distribution are unique, so  $h'(\theta)^2 v(\theta) = 1$  and  $|h'(\theta)| = v(\theta)^{-1/2}$  for every  $\theta \in I$ . The continuous function  $h'$  never vanishes on the interval  $I$ , so by the intermediate value theorem it has one sign throughout. Hence  $h' = g'$  or  $h' = -g'$  on  $I$ , and integrating gives the result. ■

Convergence in distribution does not by itself imply that the *variance* of  $g(T_m)$  is close to  $1/m$ ; that needs uniform integrability, which the theorem does not assert ([Exercise \(C2\)](#) supplies it for Poisson counts).

### Corollary 22.4.2 — The classical transformations

- (a) **Counts.** If  $Y \sim \text{Poisson}(\mu)$ , then  $2(\sqrt{Y} - \sqrt{\mu}) \rightarrow N(0, 1)$  as  $\mu \rightarrow \infty$ .
- (b) **Proportions.** If  $S \sim \text{Binomial}(m, p)$  with  $0 < p < 1$  fixed and  $\hat{p} = S/m$ , then  $2\sqrt{m} \{\arcsin \sqrt{\hat{p}} - \arcsin \sqrt{p}\} \rightarrow N(0, 1)$  as  $m \rightarrow \infty$ .
- (c) **Constant coefficient of variation.** If  $Y = \mu U$  for a positive random variable  $U$  whose distribution does not depend on  $\mu$ , then  $\text{Var}(\log Y) = \text{Var}(\log U)$  exactly, for every  $\mu$ , whenever it is finite.
- (d) **Power variance functions.** If  $v(\mu) = \mu^{2a}$  on  $(0, \infty)$ , then [Eq. 22.4.1](#) gives  $\mu^{1-a}/(1-a)$  for  $a \neq 1$  and  $\log \mu$  for  $a = 1$ . Up to an affine change this is the Box–Cox transformation with  $\lambda = 1 - a$ .

*Proof.*

- (a) The characteristic function of  $(Y - \mu)/\sqrt{\mu}$  is  $\exp\{\mu(e^{it/\sqrt{\mu}} - 1 - it/\sqrt{\mu})\}$ , which tends to  $e^{-t^2/2}$  because  $e^{iu} - 1 - iu = -u^2/2 + O(u^3)$ . By the continuity theorem of Lévy (van der Vaart 1998, chapter 2),  $(Y - \mu)/\sqrt{\mu} \rightarrow N(0, 1)$ . Now write

$$2(\sqrt{Y} - \sqrt{\mu}) = \frac{Y - \mu}{\sqrt{\mu}} \cdot \frac{2\sqrt{\mu}}{\sqrt{Y} + \sqrt{\mu}}.$$

Since  $\text{Var}(Y/\mu) = 1/\mu$ , Chebyshev's inequality gives  $Y/\mu \rightarrow 1$  in probability, so the second factor tends to 1 in probability, and Slutsky's lemma gives the limit.

- (b) By the central limit theorem  $\sqrt{m}(\hat{p} - p) \rightarrow N(0, p(1-p))$ . With  $v(p) = p(1-p)$  on  $(0, 1)$ , the derivative of  $\arcsin \sqrt{p}$  is  $\frac{1}{2}\{p(1-p)\}^{-1/2}$ , so [Eq. 22.4.1](#) gives  $g(p) = 2 \arcsin \sqrt{p}$ , and [Theorem 22.4.1\(a\)](#) applies.
- (c)  $\log Y = \log \mu + \log U$ , and adding a constant does not change a variance.
- (d) Integrate  $t^{-a}$ .

■

Part (c) is exact: for gamma data with shape  $k$ ,  $\text{Var}(\log Y)$  is the trigamma function at  $k$  whatever the mean, 0.2213 for  $k = 5$  against the first-order value  $1/k = 0.2$ . Part (d) is the rule quoted in [Section 22.1](#). A known variance function determines the power; otherwise the Box–Cox likelihood estimates one, chosen partly to stabilize the variance.

## 22.4.2 How well they work

The limits say nothing about moderate means, but the variances can be computed exactly by summing probability mass functions. [Figure 22.4.1](#) and the table show them, scaled so that the target is 1.

| Poisson mean                    |       |       |       |       |       |
|---------------------------------|-------|-------|-------|-------|-------|
| $\mu$                           | 1     | 2     | 5     | 10    | 20    |
| $4 \text{ Var } \sqrt{Y}$       | 1.609 | 1.560 | 1.144 | 1.045 | 1.020 |
| $4 \text{ Var } \sqrt{Y + 3/8}$ | 0.717 | 0.924 | 1.002 | 1.001 | 1.000 |

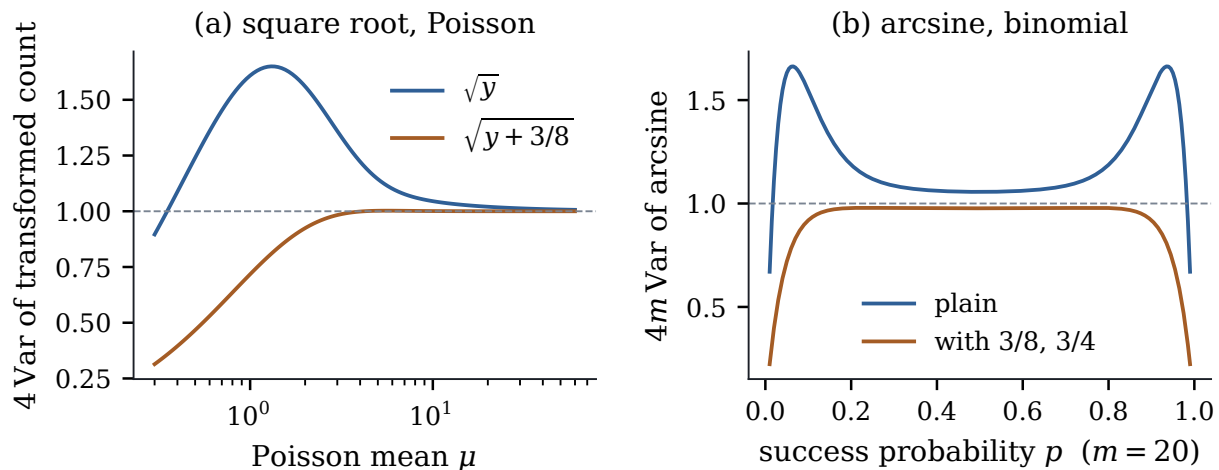


Figure 22.4.1: Exact variances of transformed counts and proportions, scaled so that the target is 1 (dashed). (a) Poisson counts:  $\sqrt{y}$  and Anscombe's  $\sqrt{y + 3/8}$ , against the mean on a log scale. (b) Binomial proportions with  $m = 20$  trials: the arcsine square root, plain and with Anscombe's constants.

Without transformation the variance of a Poisson count grows in proportion to the mean, a factor of 20 between  $\mu = 1$  and  $\mu = 20$ . After the square root the scaled variance stays between 1 and 1.7 over the same range, and from  $\mu = 5$  on it is within 15 percent of the target. Anscombe's (1948) modification  $\sqrt{Y + c} = \frac{1}{4} + (3 - 8c)/(32\mu) + O(\mu^{-2})$  (Exercise (C1)), and  $c = 3/8$  removes the  $1/\mu$  term. For binomial proportions from  $m = 20$  trials,  $4m \text{Var}(\arcsin \sqrt{\hat{p}})$  is 1.056 at  $p = 0.5$ , 1.118 at  $p = 0.25$ , 1.546 at  $p = 0.1$  and 1.632 at  $p = 0.05$ . Anscombe's version  $\arcsin \sqrt{(S + 3/8)/(m + 3/4)}$  gives 0.977, 0.979, 0.919 and 0.720. Both transformations fail near the boundary, where most of the probability sits on a few values and no smooth function can equalize anything.

```
import numpy as np
from scipy import stats

def poisson_var(h, mu):
    """Var h(Y) for Y ~ Poisson(mu), by summing the probability mass function."""
    k = np.arange(0, int(mu + 40 * np.sqrt(mu) + 60))
    p = stats.poisson.pmf(k, mu)
    m1 = p @ h(k)
    return p @ (h(k) - m1) ** 2

def binomial_var(h, m, prob):
    """Var h(S) for S ~ Binomial(m, prob)."""
    s = np.arange(m + 1)
    p = stats.binom.pmf(s, m, prob)
    m1 = p @ h(s)
    return p @ (h(s) - m1) ** 2

for mu in (1, 2, 5, 10, 20):
    print(f"Poisson mean {mu:2d}: 4 Var sqrt(Y) = {4 * poisson_var(np.sqrt, mu):.3f}")
    print(f"4 Var sqrt(Y + 3/8) = {4 * poisson_var(lambda k: np.sqrt(k + 3 / 8),
```

```

    mu):.3f}")
m = 20
for prob in (0.05, 0.1, 0.25, 0.5):
    plain = 4 * m * binomial_var(lambda s: np.arcsin(np.sqrt(s / m)), m, prob)
    anscombe = 4 * m * binomial_var(lambda s: np.arcsin(np.sqrt((s + 3 / 8) / (m + 3
        / 4))), m, prob)
    print(f"binomial m = {m}, p = {prob:.2f}: 4m Var = {plain:.3f}    with 3/8: {
        anscombe:.3f}")

```

[Run this code online](#)

A stabilized scale also gives a check: for Poisson counts the residual variance on the square-root scale should be near  $\frac{1}{4}$ . A much larger value says the counts are *overdispersed*, and then the square root no longer stabilizes ([Exercise \(B1\)](#)).

### 22.4.3 Estimating the variance function

With replicates at several settings, plot the log standard deviation of each group against the log of its mean. If  $s \propto \mu^a$  the points lie near a line of slope  $a$ , and the power  $1 - a$  stabilizes the variance ([Exercise \(B3\)](#)), an idea that goes back to Bartlett (1947). Without replicates, residuals can be grouped by fitted value, but misfit of the mean then inflates some spreads, so the plot should come *after* the mean function is roughly right. Whether the variance depends on the regressors at all is tested formally by Theorem 21.2.2.

### 22.4.4 Transform, weight, or model the variance

Transformation is one of three remedies for a variance that depends on the mean.

1. **Transform.** Fit the linear model to  $g(y)$ . The variance becomes nearly constant, but the *mean* is now modelled on the new scale, and statements about  $y$  need the retransformation of [Section 22.5](#).
2. **Weight.** Keep the scale and fit by weighted least squares with weights  $1/v(\hat{\mu}_i)$  (Theorem 21.3.1 and Definition 6.9.1), or keep ordinary least squares with heteroscedasticity-consistent standard errors (Theorem 21.4.2).
3. **Model.** Specify the mean through a link function and the variance through  $v$ : a generalized linear model, such as Poisson or logistic regression, whose coefficients refer to the mean of  $y$  itself. [Chapter 34](#) develops this modern default.

The first two can be combined. The *logit* of a binomial proportion is not variance-stabilizing, but its variance is about  $1/\{m p(1 - p)\}$ , so weighted least squares on empirical logits with weights  $m\hat{p}(1 - \hat{p})$  gives a linear model for the log odds ([Exercise \(B2\)](#)), the approach of Grizzle, Starmer and Koch (1969) for categorical data.

### 22.4.5 Exercises

#### 22.4.5.1 A. Check your understanding

**Exercise A1.**

Find the variance-stabilizing transformation for  $v(\mu) = \mu^3$ , the variance function of the inverse Gaussian distribution. Which Box–Cox power is it?

**Exercise A2.**

Let  $T_m$  be the mean of  $m$  independent exponential waiting times with mean  $\theta > 0$ . Find the variance-stabilizing transformation. Use [Theorem 22.4.1\(b\)](#) to show that the square root, although it stabilizes Poisson counts, does not stabilize  $T_m$ , and find the limiting variance of  $\sqrt{m}(\sqrt{T_m} - \sqrt{\theta})$ .

**22.4.5.2 B. Practice****Exercise B1.**

Negative binomial counts have  $v(\mu) = \mu + \mu^2/\kappa$  for a shape parameter  $\kappa > 0$ . Show that [Eq. 22.4.1](#) gives  $g(\mu) = 2\sqrt{\kappa} \sinh^{-1} \sqrt{\mu/\kappa}$ . Show that  $g$  behaves like  $2\sqrt{\mu}$  for  $\mu \ll \kappa$  and like  $\sqrt{\kappa} \log \mu$  plus a constant for  $\mu \gg \kappa$ , and interpret.

*Solution.* Substitute  $t = \kappa u^2$ :  $dt = 2\kappa u du$  and  $t + t^2/\kappa = \kappa u^2(1 + u^2)$ , so the integrand becomes  $2\kappa u du / \{\sqrt{\kappa} u \sqrt{1 + u^2}\} = 2\sqrt{\kappa} du / \sqrt{1 + u^2}$ , whose integral is  $2\sqrt{\kappa} \sinh^{-1} u$  with  $u = \sqrt{\mu/\kappa}$ . For small  $u$ ,  $\sinh^{-1} u \approx u$ , giving  $2\sqrt{\mu}$ . For large  $u$ ,  $\sinh^{-1} u = \log(2u) + o(1)$ , giving  $\sqrt{\kappa} \log \mu$  plus a constant. Small counts behave like Poisson counts and need a square root; large counts have a nearly constant coefficient of variation and need a logarithm.

**Exercise B2.**

Let  $S \sim \text{Binomial}(m, p)$  with  $0 < p < 1$ . Show by the delta method that  $\sqrt{m} \{\text{logit}(S/m) - \text{logit } p\} \rightarrow N(0, 1/\{p(1-p)\})$ , where  $\text{logit } p = \log\{p/(1-p)\}$ . For independent groups  $i = 1, \dots, k$  with  $S_i \sim \text{Binomial}(m_i, p_i)$  and  $\text{logit } p_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , write down the weighted least squares estimate of  $\boldsymbol{\beta}$  from the empirical logits, and its approximate covariance matrix.

*Solution.* The derivative of  $\text{logit } p$  is  $1/\{p(1-p)\}$ , and  $\sqrt{m}(S/m - p) \rightarrow N(0, p(1-p))$ , so the limiting variance is  $p(1-p)/\{p(1-p)\}^2$ . With  $\ell_i = \text{logit}(S_i/m_i)$  and  $\mathbf{W} = \text{diag}\{m_i \hat{p}_i(1 - \hat{p}_i)\}$ , the estimate is  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \boldsymbol{\ell}$  (Definition 6.9.1), with approximate covariance  $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ , since the weights are the approximate inverse variances and no  $\sigma^2$  needs estimating. The empirical logit is infinite when  $S_i = 0$  or  $m_i$ , which is why  $\frac{1}{2}$  is often added to  $S_i$  and to  $m_i - S_i$ .

**Exercise B3.**

Replicate observations at four settings have means 10, 40, 160, 640 and standard deviations 2, 5.7, 16, 45. Fit the spread–level line  $\log s = c + a \log \mu$  described above by eye or by least squares, and use [Corollary 22.4.2\(d\)](#) to choose the power.

## 22.4.5.3 C. Going deeper

**Exercise C1.**

Let  $Y \sim \text{Poisson}(\mu)$  and  $c \geq 0$ . Expand  $\sqrt{Y+c} = \sqrt{\mu}(1+u)^{1/2}$  with  $u = (Y-\mu+c)/\mu$  to third order in  $u$ , use the Poisson central moments  $E(Y-\mu)^2 = \mu$ ,  $E(Y-\mu)^3 = \mu$ ,  $E(Y-\mu)^4 = 3\mu^2 + \mu$ , and show formally that  $\text{Var } \sqrt{Y+c} = \frac{1}{4} + (3-8c)/(32\mu) + O(\mu^{-2})$ . (The expansion is formal: making the remainder rigorous needs a bound on the tail of  $Y$ .)

**Exercise C2.**

[Corollary 22.4.2\(a\)](#) is a statement about distributions. Show that the variance converges too:  $4 \text{Var } \sqrt{Y} \rightarrow 1$  as  $\mu \rightarrow \infty$  for  $Y \sim \text{Poisson}(\mu)$ . Use the bound  $(\sqrt{Y} - \sqrt{\mu})^2 \leq (Y-\mu)^2/\mu$ , and the fact that a family of nonnegative random variables that converges in distribution, and whose means converge to the mean of the limit, is uniformly integrable.

*Solution.* The bound holds because  $\sqrt{Y} - \sqrt{\mu} = (Y - \mu)/(\sqrt{Y} + \sqrt{\mu})$  and  $\sqrt{Y} + \sqrt{\mu} \geq \sqrt{\mu}$ . The variables  $V_\mu = (Y - \mu)^2/\mu$  converge in distribution to  $\chi^2(1)$  (by the proof of [Corollary 22.4.2\(a\)](#)) and have mean exactly 1, the mean of the limit, so they are uniformly integrable, and so is the dominated family  $W_\mu = 4(\sqrt{Y} - \sqrt{\mu})^2$ . Since  $W_\mu \rightarrow \chi^2(1)$  in distribution, uniform integrability gives  $E W_\mu \rightarrow 1$ . The same argument applied to  $2(\sqrt{Y} - \sqrt{\mu})$ , which is dominated by  $1 + W_\mu$ , gives  $E\{2(\sqrt{Y} - \sqrt{\mu})\} \rightarrow 0$ . Hence  $4 \text{Var } \sqrt{Y} = E W_\mu - [E\{2(\sqrt{Y} - \sqrt{\mu})\}]^2 \rightarrow 1$ .